

OGG Simulation - EcoBreeze

ADS4293 Professional Analytics

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Initial Decision

Our main objective is to **maximize** our overall competitive position by increasing **market share** in both leadership and focus regions, while ensuring that raw-materials usage, production capacity, and financial resources still **support strong profitability**.

To achieve this, we adopted the “**Differentiation**” strategy. This means we aim to compete through innovation and added value rather than low prices.

For the **leadership region**, we selected **Region 4**, which has the highest demand and the warmest climate, making it an attractive market to pursue market-share leadership. For our **focus region and plant location**, we chose **Region 2**. Region 2 offers lower minimum wages than Regions 1 and 4, helping us control labor costs. At the same time, it is a technological hub with high incomes and a young population, which aligns with our innovation-driven positioning and supports long-term competitiveness and profitability.

Quarter 1

According to the manual, there were fewer decision points in Q1:

1. Choice of machinery model
2. R&D investment level
3. Raw material purchase quantity
4. Packaging and parts
5. Salesforce allocation to each region
6. Wage level and benefit
7. Purchase of information reports

We expected to receive the raw material in the next quarter, and the manufacturing and sales would begin from Q2. No revenue was generated.

In line with the “**Differentiation**” strategy, we choose **premium packaging and higher-quality materials**, combined with a **significant investment in Research & Development and training**, to continuously improve product and employees’ performance and raise our level of innovation over time. To execute this strategy, our group divided responsibilities across four key business functions: Marketing & Sales, Human Resources, Finance, and Operations.

For **Operations**, we selected **Machinery Model 2**, which is **more expensive but provides higher initial productivity**. Without knowing the exact impact of R&D quantity on innovation, we chose Innovation Level 2 as a starting point. **To ensure we have enough raw material for manufacturing later, we purchased the maximum quantity of 1,200 units**. We did not fully consider inventory costs at this stage, but 1,200 seemed a simple and “safe” choice to secure supply. We also chose premium packaging and parts to remain consistent with our Differentiation strategy.

For **Marketing & Sales**, the only decision was the **allocation of salespeople**. We placed more salespeople in Regions 2 and 4 (our focus and leadership regions) and fewer in Regions 1 and 3 to align with our regional priorities.

In **Human Resources**, we wanted to **make employees satisfied**. So we set wages slightly above the minimum. And we added four benefits —Tuition Fee, Retirement Plans, Medical Care, and Life Insurance —assuming that Region 2 has a university and that some employees might be students who would value these benefits.

In **Finance**, we purchased only one **demand forecast report**. Since there was no revenue in Quarter 1, we intentionally constrained this spending and decided to observe the first cycle of operations before investing more in reports.

Quarter 2

Quarter 2 was our first selling quarter, and the key decisions were to set **prices and allocate units** to each region. Given our productivity, we used the maximum production capacity and then distributed units across regions.

We found pricing especially difficult because of the **pricing policy** (a penalty if our price is more than 30% above the market level), so we **purchased an additional report with competitors' sales prices**. The report showed that one competitor, LOLI Climatizer's INC., was pricing its products extremely low compared to others. This suggested that not only our team but possibly several teams miscalculated the cost, which increased our concern about potential market penalties.

However, when we calculated our own prices for this quarter, we still made mistakes in our internal logic. Our result of **COGS** and freight cost did not match with the OGG calculation, and our prices were not based on an accurate breakeven analysis. We also relied heavily on the preview income statement, which assumes that all planned units are sold. Because this was our first selling quarter, we had no real idea of the conversion rate, and the actual sales ended up far below the planned volume. As a result, our conversion rate was very low, and we likely triggered the penalty for being overpriced in some regions.

Overall, Quarter 2 showed that we need a more structured decision-support tool, and we need to try to find a way to let the other team recalculate and raise their price.

Quarter 3

In Quarter 3, we focused on addressing the problems identified in Q2, particularly pricing, conversion rates, and profit estimation. Because the OGG profit preview is based on planned sales rather than actual sales, it became less useful once the gap between planned and actual units was large. To improve our decisions-making, **we rebuilt the decision logic in Excel, so we could estimate profit using our own assumed conversion rate and different price scenarios**.

We also confirmed that our earlier price calculations were not accurate, partly because our understanding of COGS did not fully match the simulator's calculation. Under the Differentiation strategy, our use of high-quality materials and R&D investment keeps costs high, **so we have limited room to lower price without getting a penalty from the pricing policy**. At the same time, we need a relatively high conversion rate to cover our initial investments. This tension, combined with some competitors choosing

very low margins, left our cash flow under pressure, and we had to take an additional loan of about 300,000 in Quarter 3.

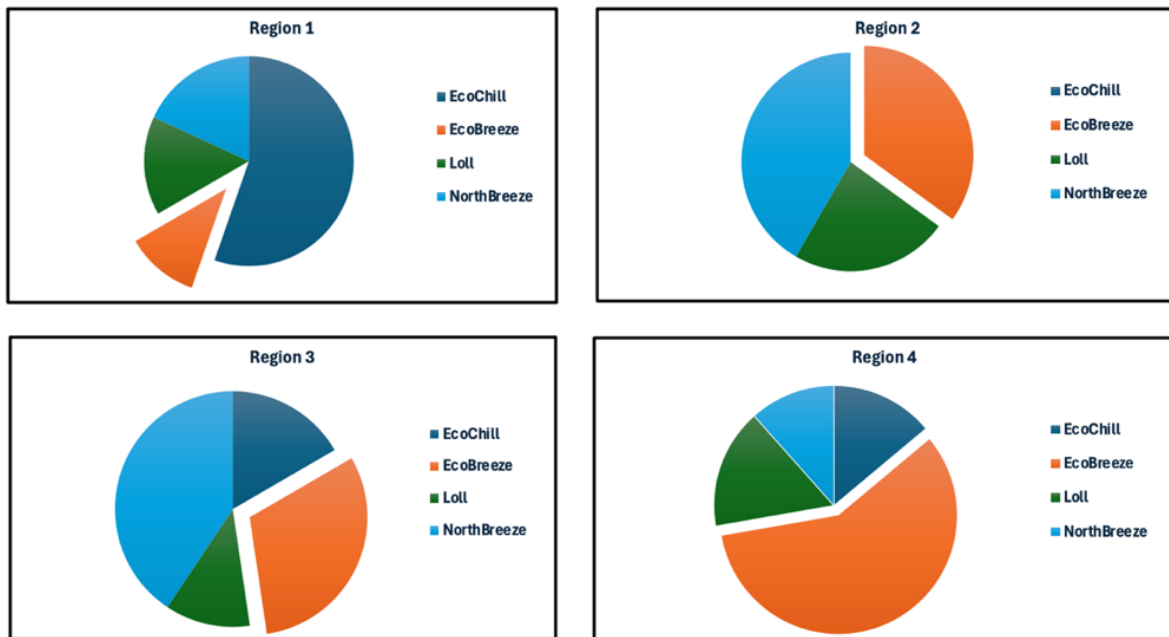
To support sales in the short term, we decided to **increase the sales commission rate**. Although this reduces the per-unit margin, **we expect a higher commission to motivate the salesforce**, improve conversion rates, and increase overall revenue more than a lower commission would.

Quarter 4

In Quarter 4, we assumed demand would not be as strong as in previous quarters, primarily due to **seasonality** and weather changes. As a result, we **anticipated lower sales** and fewer units to be sold. The calculated conversion rate of 52% ultimately left us with a high level of finished goods inventory. This low conversion rate, especially in regions with **weaker A&P rankings**, highlighted the need for more targeted marketing and stronger competitiveness in focus markets.

Our negative net profit in previous quarters led to significant cash-flow constraints. With revenue failing short of expectations while costs remained high, our operating cash flows declined sharply. To maintain operations and cover our expenses, we were forced to rely heavily on automatic loans throughout the quarter. This reliance on external financing reflects pressure on our liquidity and indicates that our current sales performance is insufficient to meet our cash requirements. Additionally, our level of innovation remained low, which affected our ability to differentiate the product and strengthen our market position.

Team satisfaction in Quarter 4 remained neutral. To **improve satisfaction** and support stronger workplace motivation, we plan to reevaluate our HR structure in the upcoming quarter. This may include increasing salaries, investing more in training, or enhancing fringe benefits to ensure employees feel supported and engaged.



Quarter 5

Entering Quarter 5, our team identified an important new insight: **sales units are significantly influenced by the relative rankings of Advertising & Promotion investments across regions**. This discovery shifted the way we approached our marketing allocation for the quarter. In Region 2, where competitive pressure was rising, we decided to increase our A&P investment to enhance our visibility and strengthen brand positioning. Additionally, we allocated an extra salesperson to further support our demand-generation efforts and lowered our price in this focus region to remain competitive and stimulate higher sales volume.

We aimed to secure the top A&P ranking in Regions 1, 2, and 4 to protect and grow our market share. In **Region 3**, we noticed competitors lowering prices and competing heavily for market share. Rather than engaging in this price war, **we stepped back from this region** and concentrated our marketing efforts in the other regions where our differentiation strategy could generate stronger returns.

We **adjusted the benefits and increased the salary** according to the information given to us in the **newsletter**. We invested in both training and CSR initiatives. With these combined efforts, we finally achieved a higher level of team satisfaction, which supports stronger workforce motivation and aligns with our goal of maintaining a well-supported, high-performing team. We also **increased our R&D** investment to elevate our level of innovation, which is an essential component of our differentiation strategy. Strengthening product quality and innovation ensures long-term competitiveness.

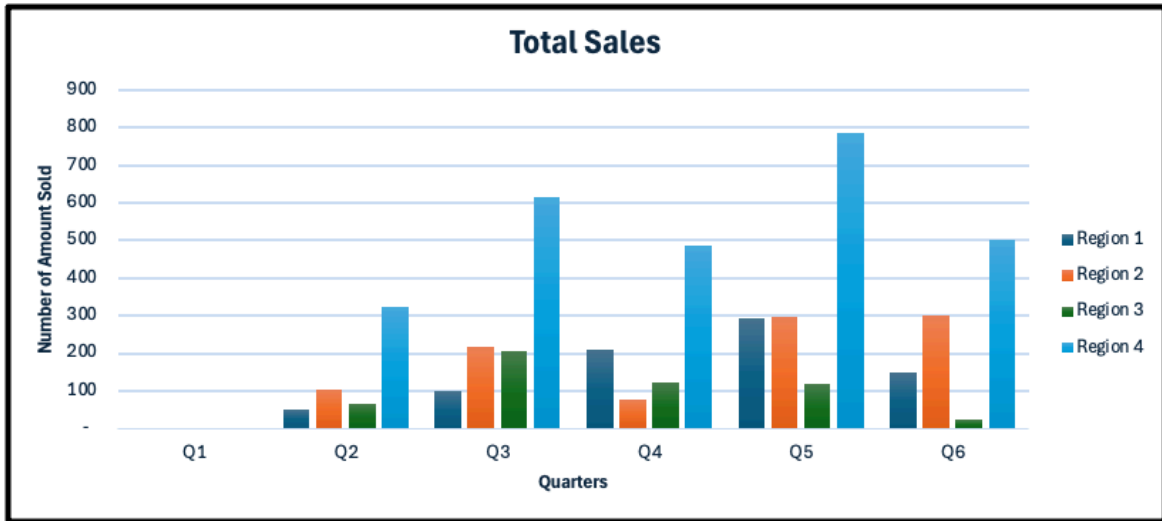
When setting our price, we communicated with competitors (LOLL & EcoChill) and learned that both planned to increase their prices. Using this information, we selected a safe upper-bound pricing level that kept us competitive while remaining profitable.

Quarter 6

From a financial perspective, sales income reached \$2,554,023, which, although lower than the high spike in Quarter 5, remains a major improvement over the previous quarter's results and reflects strong market performance. Our cost of goods decreased significantly, resulting in a healthier gross margin of \$1,557,964. With a generated profit of \$363,502, **our company shows a clear recovery from the losses seen in past quarters**. This improvement indicates that our pricing decisions, A&P direction, and product position are beginning to produce sustainable financial benefits.

Operationally, we ended with a **productivity rate of 91%**. Our inventory value demonstrates that we purchased enough raw material to avoid shortages without over-investing. We believe maintaining higher inventory volume remains appropriate for supporting sales in upcoming periods. Our high conversion rate confirms that our consistent A&P spending and competitive pricing continued to strengthen our market share in Regions 1, 2 and 4. Even though Region 3 remained weak due to strong competitive and lower A&P ranking, the strong performance in the other three regions has well compensated for this gap.

Employee satisfaction remained steady for this quarter, as we maintained the same HR structure and investment levels that previously contributed to strong team morale. Maintaining this stability positions us well for the upcoming quarter, ensuring that our team remains motivated and aligned with our operational goals.



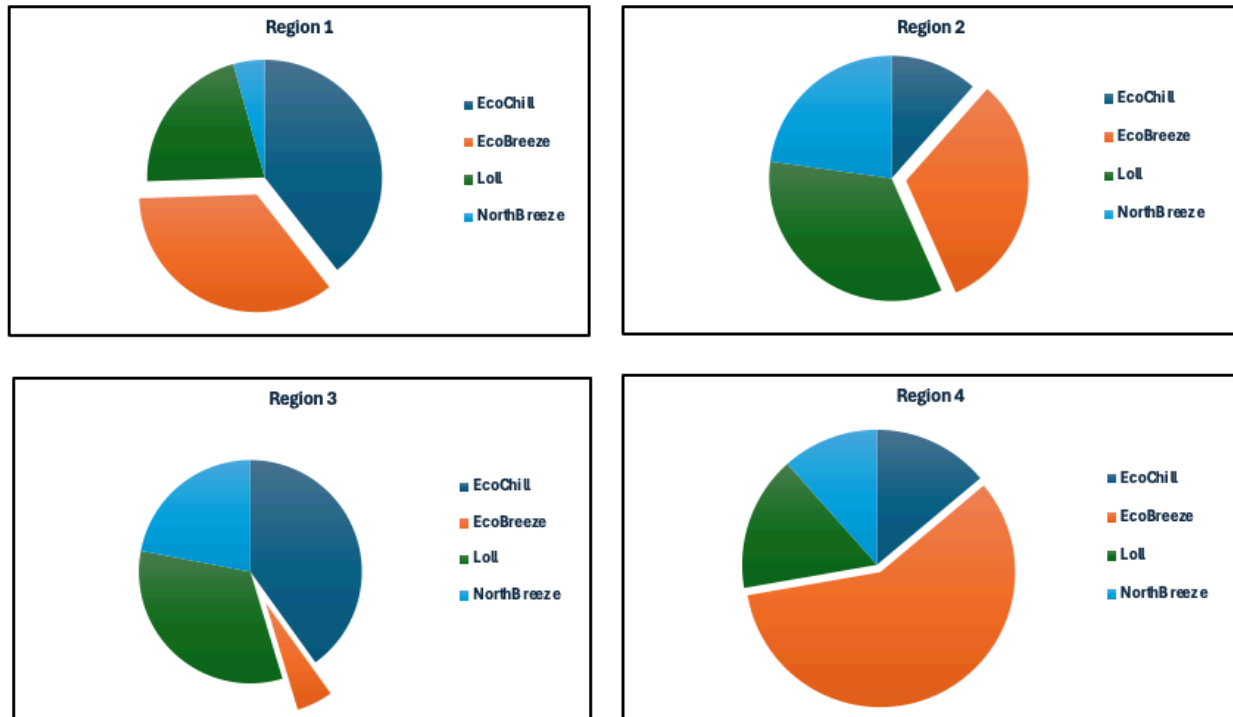
Quarter 7

After experiencing a high conversion rate in the previous 2 quarters and consistently reporting positive net profit, we **decided to push overtime to increase market share**. And we continue increasing our price, because of the higher cost over time.

We also made more adjustments to the distribution, with **more distribution in region 1 and less in region 2**. Because we have the lower price set in region 2, combined with lower demand there than in region 1, selling more to region 1 might benefit us more, both in market share and revenue.

The result for the quarter was ideal, and we sold the highest number of units ever, 1284 pieces. However, lower net profit compared to Q6 due to higher fixed costs.

Actual S.C.R	82%	85%	91%	92%	
Region	1	2	3	4	Total
Unit sale price	\$ 2,899.00	\$ 2,399.00	\$ 2,899.00	\$ 2,899.00	/
Planned Sale	439	249	43	732	1,600
Actual Sale	359	212	39	674	1284
Advertisement unit	7	9	1	7	
A&P Investment	\$ 89,086.27	\$ 53,761.59	\$ 1,246.57	\$ 148,544.76	\$ 292,639.19
New Hired sales	1	0	-1	0	/
Active Sales Force	3	3	1	4	11
Total Region Demand					From Report
Market Share by Region					From Report



Quarter 8

We relied heavily on overtime throughout the quarter, which increased our labour costs and reduced efficiency. In addition, we rushed several key decisions and moved away from our initial pricing analysis. Instead of maintaining our planned prices of \$2,899 for Regions 1, 3 and 4 and \$2,499 for Region 2, we increased \$2,899 up to \$3,000.

This hasty change negatively affected demand and contributed to our loss. As a result, we ended the quarter with 584 units of finished goods inventory, showing that our production levels exceeded actual sales. Furthermore, it decreased our profit margins, contributed to our overall loss, and was reflected in a lower conversion rate. Despite these setbacks, our HR decisions remained strong, we were satisfied with the structure we had in place and chose to maintain the same approach as in previous quarters.

On a positive note, our innovation score increased to three, highlighting our continued commitment to product development and differentiation.

Quarter 9

This was a quarter of recovery and recalibration. Following our not ideal Quarter 8 where a rash last-minute decision to boost unit prices to \$2,999 in Regions 1, 3, and 4. This led us to experience our lowest conversion rate of 62% since quarter 4. Due to this we returned all settings to our original levels to prepare for a conservative quarter to stabilize our performance.

We continued our R&D investment reaching level 3 innovation this quarter. We've held the R&D Quantity high and steady at level 9 since Quarter 5 as it is very important to our Differentiation Strategy to continue innovating.

During this the decision for this quarter we also reconsidered the benefits and fallbacks of over time. Which we began exercising in Quarter 7 to gain a higher market share in our leadership and focus region.

We compared our maximum capacity using overtime and a conservative level of production to evaluate whether or not we should continue overtime production this quarter or halt it as we pulled back and readjusted other metrics after our unit price miscalculation in Quarter 8.

In this Quarter we also began increasing our **CSR investment to Maximum Capacity of 9** as we revisited the manual seeking advice how to increase our productivity. We found that investment in CSR will increase productivity and in turn, our conversion rate. This decision was very low risk as it only was an \$8,000 bet to increase our current level of 5 to the max level of 9.

Quarter 10

In Quarter 10, we continued to rely on employee overtime this quarter, but despite the increased labour strain, our overall performance improved substantially compared to the previous period. We sold significantly more units, generating an impressive sales income of \$4.5 million. Our strategic decisions also resulted in a strong ending cash balance of over \$1.5 million, nearly a \$900,000 increase from Quarter 9, demonstrating financial stability.

With a slight increase in our conversion rate and 427 units of finished goods still in inventory, we anticipate selling these next quarter as we are anticipating a higher demand for the following quarter.

Our HR decisions remained consistent, maintaining stable salaries, training levels, and benefits to support employee morale and ensure that our workforce continues to feel motivated and aligned with our operational goals.

Part of our Financial decisions is purchasing reports to view our competitors progress throughout the Quarters, we have not purchased any reports for this quarter, however, we will ensure to purchase the necessary reports needed in order for us to make more accurate decisions and achieve a high profit and lead the regions.

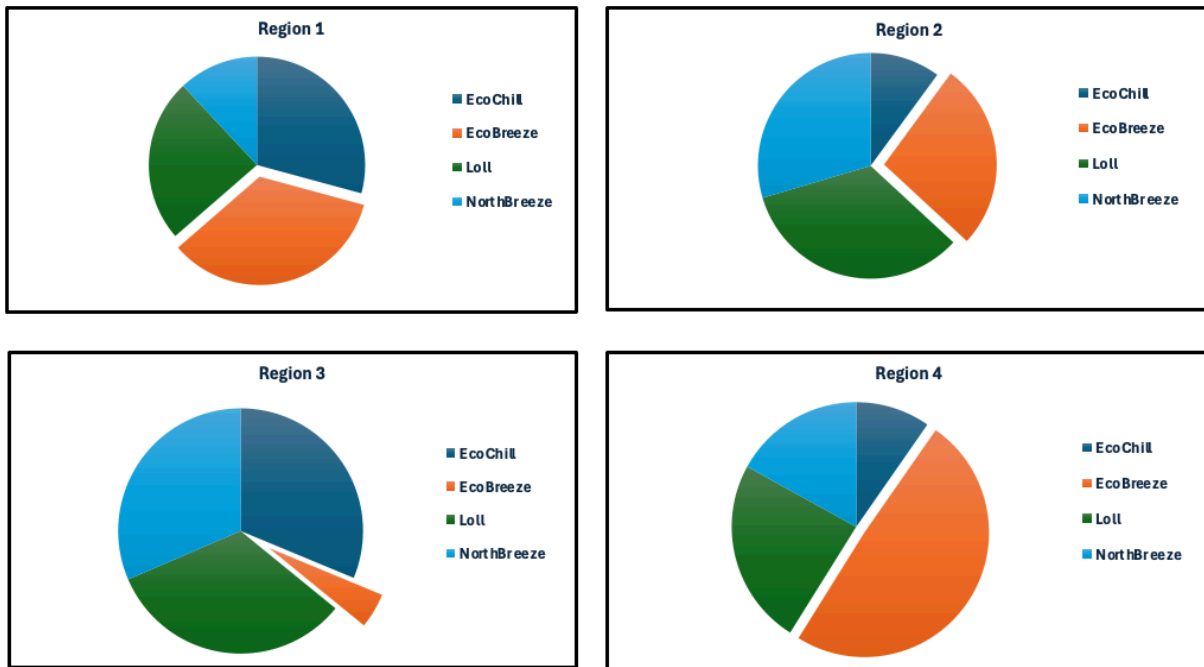
Quarter 11

In Quarter 11, recognizing the increased demand associated with the assumed to be Summer period, we maximized our production capacity by utilizing overtime. As a result, we manufactured 1,513 units and sold 1,850 units, generating strong sales revenue of \$5.2 million.

With an optimized price variance, we achieved an exceptional conversion rate of 96% and secured the highest market shares in Regions 1, 2 and 4, consistent with our strategic objectives.

Our HR decisions stayed stable as the employees were satisfied and showed increased productivity.

Throughout the quarters, we closely monitored market share trends and competitor pricing to validate the impact of our strategic decisions and ensure we continued control over price variance.



Quarter 12

The previous quarter reflected a continuation of our earlier optimized performance, with production maintained at its peak capacity. Although we anticipated a slight decline in sales due to anticipated colder season, our performance remained strong. Despite these seasonal expectations, we achieved an 83% conversion rate and generated \$3.8 million in sales revenue.

By the end of the quarter, our accumulated profit reached approximately \$2.2 million, with no outstanding debt. We continued to dominate the market share, supported by our product quality that remained relatively high among all competitors.

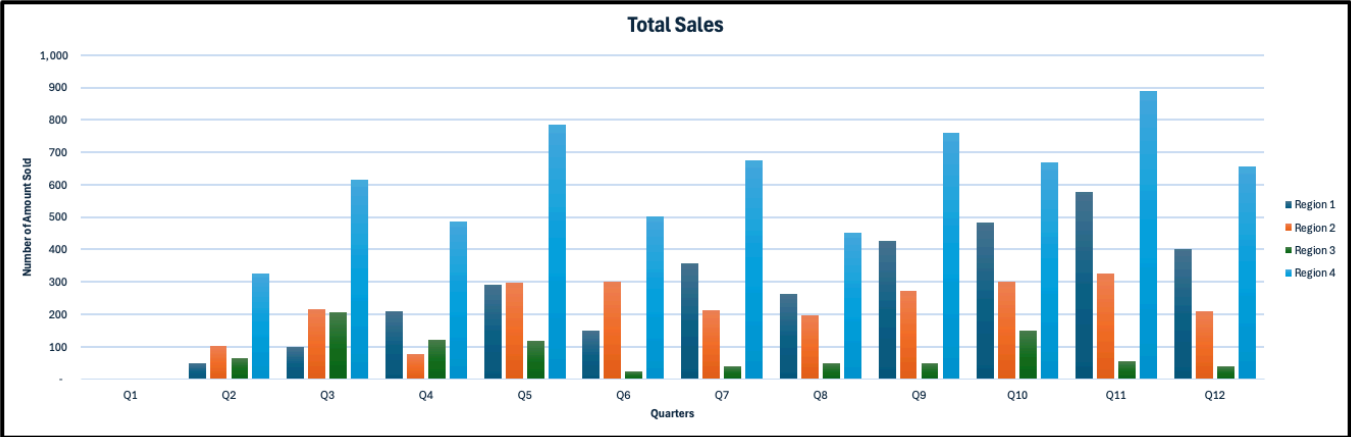
Ending

Throughout this simulation game, we realized that the key to achieving your end goal is ensuring your decisions align with your strategy.

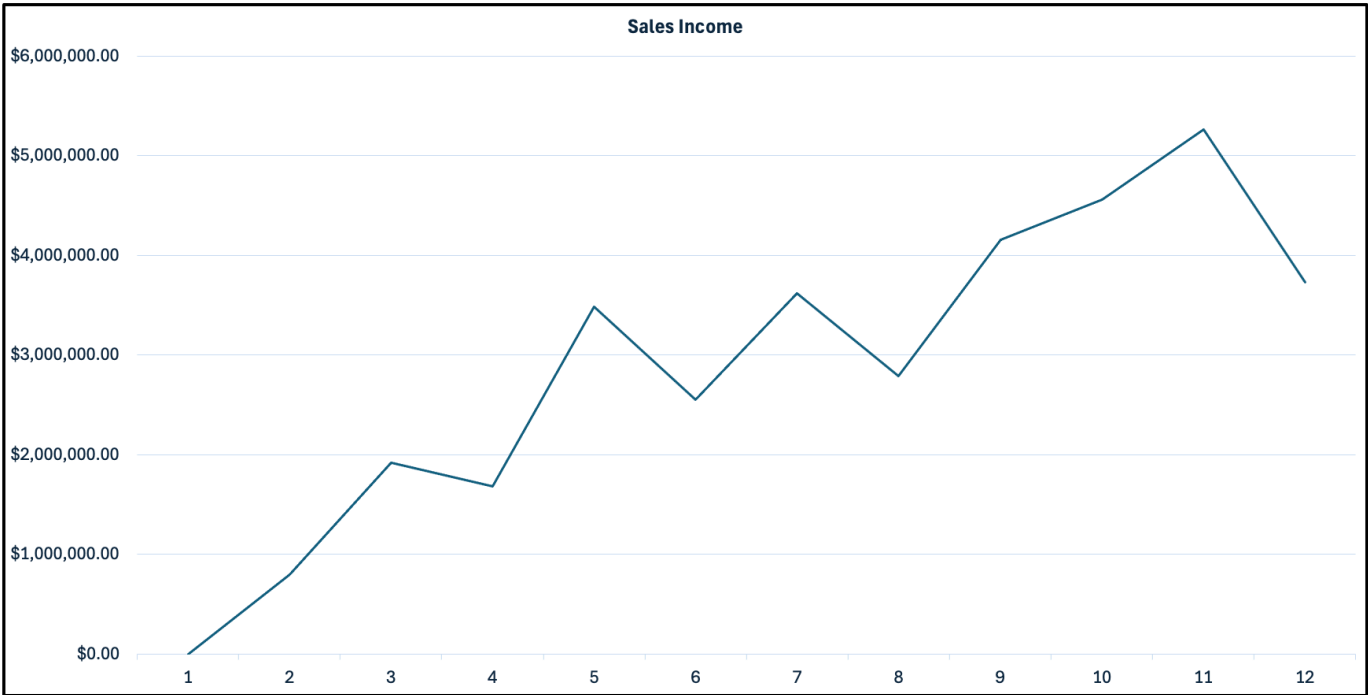
Correct pricing, supported by our breakeven analysis, was critical when launching in this simulation. We felt it was important to consider our competitors' pricing points, as their decisions would directly influence our performance. After reviewing a few rounds of reports that we purchased throughout the quarters, we were able to refine our pricing decisions and select the price variance that produced the highest conversion rate.

Our decisions were driven by our target outcomes and strategic direction. With only four teams participating in the simulation, each group's choices had the potential to influence the entire market, prompting us to adjust our decisions carefully each quarter.

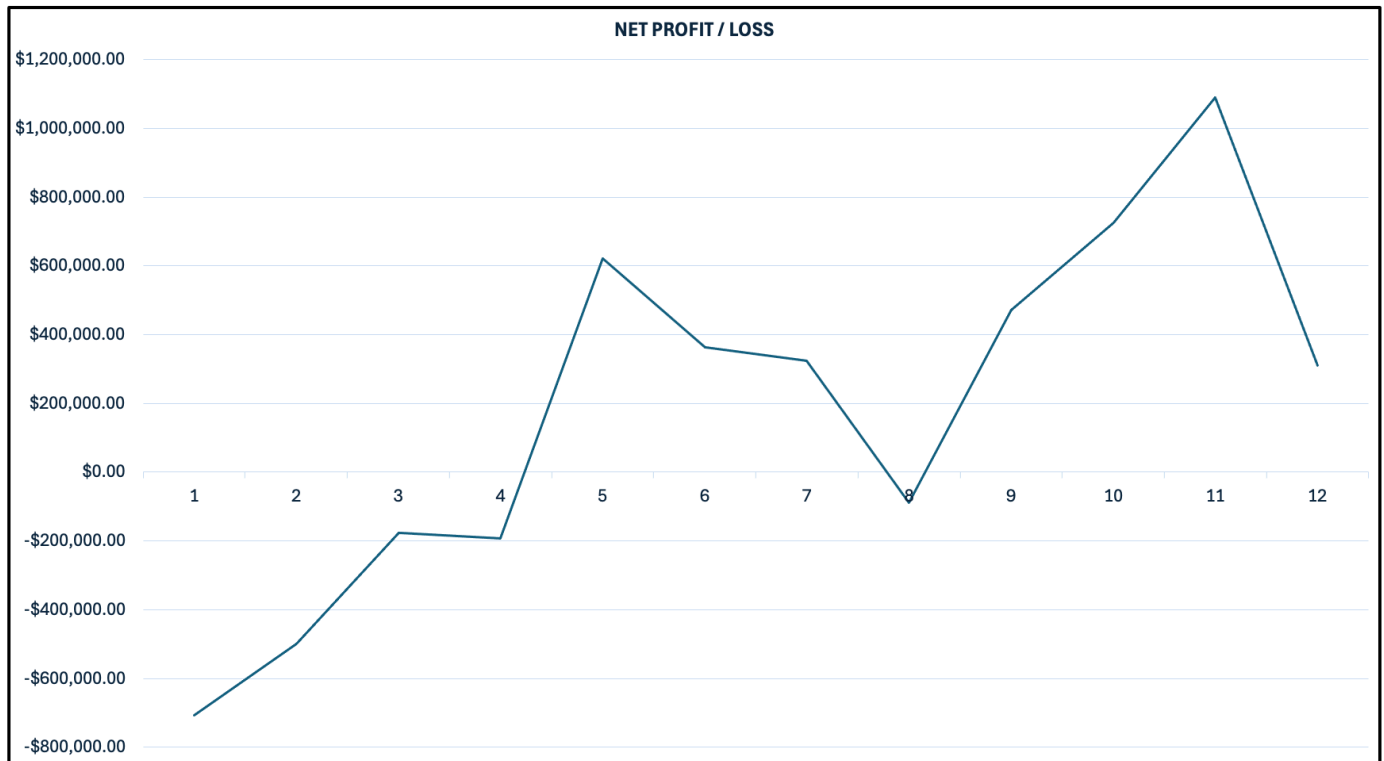
Setting the appropriate price proved to be essential, especially since initial pricing affected the pace of market expansion in the early rounds. This experience underscored the value of breakeven analysis and deliberate price setting. In addition, the quarterly information reports were vital in guiding our decisions as the simulation progressed. We did not describe the model in detail in this report, our technical details and calculations are documented in the manual and spreadsheet.



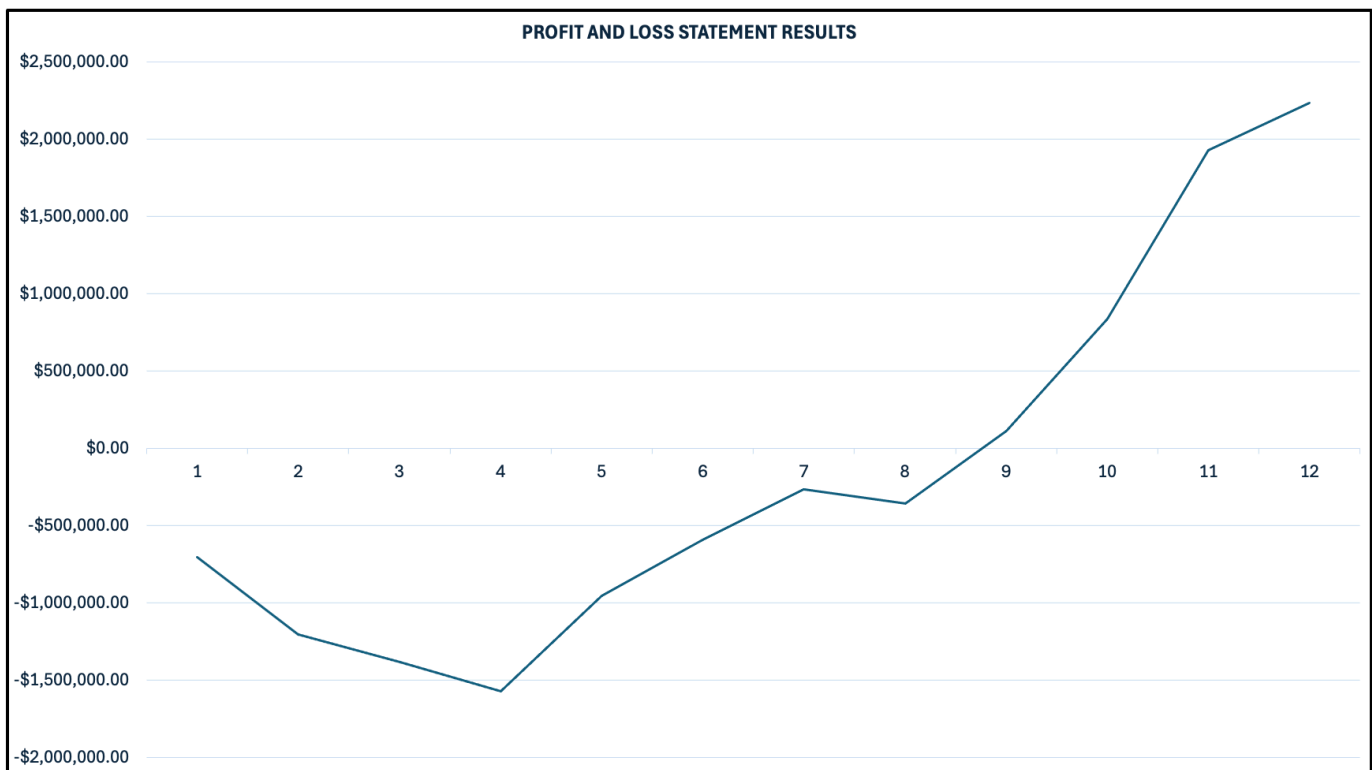
Final Total Sales Per Quarter.



Sales Income Results Per Quarter.



Net Profit (Loss) Results Per Quarter.



Profit and Loss Statement Results